

Montazar L. Matlih

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EDUCATION

National University (NU-D)

Bachelor of Science in Computer Science, Specialization in Machine Learning (BSCS-ML)

Cavite, Philippines

Expected 2027

- **Academic Honors & Recognition:** 1st Place Champion, Project IMPACT; DOST Calabarzon & DSWD Endorsee; 3× Invited STEM Speaker.
- **Relevant Coursework:** Edge Computer Vision, Multi-Agent Systems, Deep Learning & Neural Networks, Natural Language Processing, Cryptography & Systems Security, Distributed Systems, Data Structures & Algorithms.

TECHNICAL SKILLS

Programming Languages: Python, Rust, TypeScript, JavaScript, SQL, C/C++, Bash

AI & Machine Learning: Multi-Agent LLM Orchestration (LangGraph), Vision-Language Models (Qwen2-VL, Qwen-VL), Model Compression (INT8 Dynamic Quantization), Foundation Models (IBM/NASA Prithvi-100M), Edge Computer Vision (YOLO11, PatchCore), Multimodal Fine-Tuning (LoRA), OpenCV, XGBoost, Transformers.js

Edge & Systems Engineering: Bare-metal GPU Provisioning (AMD Instinct MI300X, ROCm 6.0), LLM Serving (vLLM), Edge Device Inference, FastAPI, WebSockets, Linux/Unix, Embedded Systems

UI & UX Architecture: Tauri (v2), React, Vite, TypeScript, Tailwind CSS, REST APIs, Vercel Serverless, Web Audio API

Security Architecture: Argon2id Key Derivation, ChaCha20-Poly1305 AEAD, Steganography (LSB/EOF file injection), Zero-Knowledge Vault Architecture, BIP-39 Passphrase Standards

PROFESSIONAL & ENTREPRENEURIAL EXPERIENCE

ChipSentinel

Founder & Lead AI Architect

Cavite, Philippines

May 2026 – Present

- Founded and architected an industrial edge anomaly detection platform for semiconductor and PCB manufacturing lines to intercept micro-defects before packaging.
- Implemented a two-stage hybrid inspection architecture combining PatchCore for ultra-fast edge anomaly localization with Qwen-VL for contextual scene-level defect reasoning.
- Provisioned and benchmarked on bare-metal AMD Instinct MI300X (192GB HBM3) running ROCm 6.0, serving high-throughput real-time inference via FastAPI and an interactive React monitoring console.

Blacksite Node

Lead Security Systems Developer

Open-Source

May 2026 – Present

- Engineered a zero-knowledge local password vault and encrypted notepad in Rust and Tauri v2 with strict memory-hard constraints, local-first isolation, and zero-trust IPC boundaries.
- Implemented defense-in-depth cryptographic primitives including Argon2id key derivation, ChaCha20-Poly1305 AEAD authenticated encryption, and an emergency cryptographic duress protocol.
- Integrated an NLP-based password entropy evaluator moving beyond traditional rule-based meters, dual LSB/EOF steganographic export, and multi-lingual BIP-39 mnemonic generation across 12,288 words in 6 languages.

Prod

Systems Developer & FOSS Author

Open-Source

July 2026 – Present

- Designed and published a high-performance minimalist desktop productivity timer enforcing the 90/15 Ultradian rhythm with near-zero CPU and memory footprint.
- Developed a native Rust backend utilizing Tauri IPC for persistent multi-monitor window state memory, hardware-accelerated 60fps animations, and custom-synthesized Web Audio API alert chimes.

FLAGSHIP TECHNICAL PROJECTS

ShellWise Smart Bin — Edge Waste Classifier (Project IMPACT)

DOST & DSWD Endorsed

Lead ML Engineer | NU-D × REZBIN Hackathon

April 2026

- Won **1st Place** at NU-D × REZBIN Hackathon; earned official endorsement from the Department of Science and Technology (DOST Calabarzon) and DSWD for regional public sector edge-AI deployment.
- Engineered and deployed an ultra-compact YOLO11n object classification pipeline optimized for resource-constrained edge hardware for sub-second automated waste sorting.

Project ARK — Autonomous Reconnaissance Kinematics

Global Deployment

AI Systems Architect | AMD Developer Hackathon

May 2026

- Engineered an agentic satellite disaster-response pipeline compressing emergency damage assessment turnaround from 72–96 manual hours to **under 60 seconds**.
- Orchestrated LangGraph multi-agent workflows routing optical telemetry from ESA Sentinel-2 L2A and NASA EONET through IBM/NASA Prithvi-100M geospatial foundation models and Qwen-VL.
- Executed inference on bare-metal AMD Instinct MI300X / ROCm 6.0; coupled geospatial feature embeddings with XGBoost classifiers to output immediate tactical relief directives.

Ocular Sentinel — Autonomous C4ISR Security System

AMD Hackathon: ACT II

Distributed Systems Engineer

July 2026

- Designed a distributed edge-to-cloud tactical surveillance architecture bridging edge detection tripwires with deep cloud multimodal vision reasoning.
- Deployed lightweight YOLO11n on edge nodes with circular frame buffers, streaming triggered anomalous timelapses over WebSockets to a ROCm-accelerated Qwen2-VL model on vLLM for zero-shot tactical incident reporting.

TSEK — Real-Time Cross-Lingual Fact-Checking System

Chrome Web Store

Lead Full-Stack ML Engineer | UNESCO Youth Hackathon

July 2026

- Built a cross-lingual, privacy-preserving Chrome extension and web application delivering real-time claim verification with **92.2% peak accuracy**.
- Applied 8-bit dynamic quantization (INT8) to mDeBERTa-v3-base, compressing model footprint by **75% (1.12 GB → ~280 MB)** for client-side in-browser inference via Transformers.js.
- Deployed a client-side gatekeeper discarding conversational noise locally, selectively querying Vercel Serverless and Gemini API only for verifiable empirical statements.

Aerial Vehicle Detection Pipeline — 94.2% mAP@50

Research Publication

Lead Computer Vision Researcher | Academic Capstone

June 2025

- Trained and evaluated an aerial-perspective computer vision pipeline achieving **94.2% mAP@50** across 9 fine-grained vehicle classes on a custom annotated dataset.
- Authored and published technical research report documenting architecture, ablation benchmarks, and optimization strategies on ResearchGate (DOI: 10.13140/RG.2.2.31747.46888).

HONORS, AWARDS & SPEAKING ENGAGEMENTS

- 1st Place Champion:** Project IMPACT — NU-D × REZBIN Hackathon for edge-deployed waste classification (Apr 2026).
- Regional Deployment Endorsement:** Department of Science and Technology (DOST Calabarzon) & DSWD (Apr 2026).
- Finalist & Contributor:** AMD Developer Hackathons (Project ARK, Ocular Sentinel) — Bare-metal AMD MI300X / ROCm (May & Jul 2026).
- Competitor:** UNESCO Youth Hackathon 2026 — TSEK Smart Credibility Companion (Jul 2026).
- 3× Invited Technical Speaker:** National University Dasmariñas STEM Career Fair (A.Y. 2024–25, 2025–26) & Campus Tour (A.Y. 2025–26).
 - Live Technical Demonstration:* JARVIS Vision — Voice-activated multimodal edge agent with YOLO11n & Edge TTS.
 - Live Technical Demonstration:* Cultural Garment Detection — Real-time traditional apparel identification.